# 🚀 Quick Reference - Supabase Edition

## ⚡ Quick Start (3 Steps)

```bash
# 1. Install
pip install -r requirements.txt

# 2. Setup API keys
cp .env.example .env
# Edit .env: Add GEMINI_API_KEY (required) + SUPABASE credentials (optional)

# 3. Run
streamlit run app.py
```

---

## 🔑 API Keys You Need

### Gemini (Required)

```bash
# Get from: https://makersuite.google.com/app/apikey
GEMINI_API_KEY=AIza...
```

### Supabase (Recommended)

```bash
# Get from: https://supabase.com/dashboard → Project Settings → API
SUPABASE_URL=https://xxx.supabase.co
SUPABASE_KEY=eyJhbGc...
```

---

## 📁 File Structure

```
task2-user-dashboard/
├── 🎨 Frontend
│   ├── app.py              # Main Streamlit app
```

```
│   └── .streamlit/config.toml    # UI theme
│
├── 🤖 AI & Database
│   ├── llm_client.py           # Gemini integration
│   ├── prompts.py              # AI prompts
│   ├── supabase_client.py      # Database client
│   └── utils.py                # Storage + validation
│
├── 📋 Config
│   ├── requirements.txt        # Dependencies
│   ├── .env                    # Your API keys (NOT in Git)
│   └── .gitignore              # Git exclusions
│
├── 🧪 Testing
│   ├── test_setup.py           # Verify setup
│   ├── quick_start.sh          # Auto setup
│   └── run.sh                  # Quick run
│
└── 📚 Docs
    ├── README.md               # Main docs
    ├── SUPABASE_SETUP.md       # Database setup
    ├── DEPLOYMENT.md           # Deploy guide
    ├── PROJECT_SUMMARY.md      # Overview
    └── This file
```

**Total: 16 files**

---

## 🗄️ Supabase Quick Setup

```bash
```

```
# 1. Create account
https://supabase.com → Sign Up

# 2. Create project
Dashboard → New Project → ai-feedback-system

# 3. Run SQL (in SQL Editor)
CREATE TABLE feedback (
  id BIGSERIAL PRIMARY KEY,
  timestamp TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  rating INTEGER NOT NULL CHECK (rating >= 1 AND rating <= 5),
  review TEXT NOT NULL,
  ai_response TEXT NOT NULL,
  summary TEXT NOT NULL,
  recommended_actions TEXT[] NOT NULL,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

# 4. Get credentials
Settings → API → Copy URL + anon key

# 5. Add to .env
SUPABASE_URL=...
SUPABASE_KEY=...
```

**Full guide:** SUPABASE_SETUP.md

---

## 🧪 Testing Commands

```bash
```

```
# Full verification
python test_setup.py

# Test Supabase only
python supabase_client.py

# Test Gemini only
python llm_client.py

# Test prompts
python prompts.py

# Test utils
python utils.py
```

---

## 🚀 Deployment (Streamlit Cloud)

```bash
# 1. Push to GitHub
git init
git add .
git commit -m "Initial commit"
git push origin main

# 2. Deploy
https://share.streamlit.io → New app → Select repo

# 3. Add Secrets (Settings → Secrets)
GEMINI_API_KEY = "your_key"
SUPABASE_URL = "your_url"
SUPABASE_KEY = "your_key"

# Done! 🎉
```

---

## 💡 Common Commands

### Run App

```bash
streamlit run app.py
```

### Install Dependencies

```bash
pip install -r requirements.txt
```

### Update Dependencies

```bash
pip freeze > requirements.txt
```

### Test Connection

```bash
python -c "from supabase_client import test_supabase_connection; test_supabase_connection()"
```

### Check Storage Mode

```bash
python -c "from utils import USE_SUPABASE; print('Supabase' if USE_SUPABASE else 'CSV')"
```

---

## 🔍 Troubleshooting

### App won't start

```bash
python test_setup.py  # Check all dependencies
```

### "API key not found"

```bash
# Check .env exists
ls -la .env

# Check key is set
python -c "import os; from dotenv import load_dotenv; load_dotenv(); print(os.getenv('GEMINI_API_KEY'))"
```

### Supabase not connecting

```bash
# Test connection
python supabase_client.py

# Check credentials
python -c "import os; from dotenv import load_dotenv; load_dotenv(); print(os.getenv('SUPABASE_URL'))"
```

### CSV data not showing

```bash
# Check CSV exists
ls -la feedback_data.csv

# View CSV
cat feedback_data.csv
```

---

## 📊 Key Functions

### Save Feedback

```python
from utils import save_feedback
save_feedback(timestamp, rating, review, ai_response, summary, actions)
```

### Load Recent

```python
```

```
from utils import load_recent_feedback
df = load_recent_feedback(5)  # Last 5 entries
```

### Get Statistics

```python
from utils import get_feedback_stats
stats = get_feedback_stats()
print(f"Total: {stats['total_count']}")
print(f"Average: {stats['average_rating']}")
```

### Search (Supabase only)

```python
from supabase_client import search_feedback
results = search_feedback("food")
```

### Filter by Rating (Supabase only)

```python
from supabase_client import get_feedback_by_rating
five_stars = get_feedback_by_rating(5)
```

---

## 🎨 Color Scheme

```css
Primary:    #6B8E23  /* Sage Green */
Secondary:  #36454F  /* Charcoal */
Accent:     #FFC72C  /* Gold */
Background: #FAF9F6  /* Cream */
Border:     #E7E5DF  /* Tan */
```

---

## 📝 Environment Variables

```bash
# Required
GEMINI_API_KEY=your_key_here

# Optional (but recommended)
SUPABASE_URL=https://xxx.supabase.co
SUPABASE_KEY=eyJhbGc...
```

---

## 🔄 Storage Modes

| Mode | When | Data Location |
|---|---|---|
| **Supabase + CSV** | Both configured | Supabase (primary) + CSV (backup) |
| **CSV Only** | No Supabase | CSV only (local file) |
| **Auto Fallback** | Supabase fails | Switches to CSV automatically |

---

## 📦 Dependencies

```txt
streamlit==1.31.0         # Web framework
pandas==2.1.4             # Data handling
google-generativeai==0.3.2 # Gemini AI
supabase==2.3.0           # Database
python-dotenv==1.0.0      # Environment vars
```

---

## 🎯 Assignment Checklist

- [ ] All code files created
- [ ] Gemini API key configured
- [ ] Supabase setup (optional but recommended)

- [ ] `python test_setup.py` passes
- [ ] App runs locally
- [ ] Pushed to GitHub
- [ ] Deployed to Streamlit Cloud
- [ ] Secrets configured in Streamlit
- [ ] Public URL works
- [ ] Screenshots taken

---

## 🆘 Quick Help

| Issue | Solution |
|---|---|
| Import errors | `pip install -r requirements.txt` |
| API key error | Check `.env` file |
| Supabase error | It's optional, app works with CSV |
| App won't start | `python test_setup.py` |
| Data not saving | Check storage mode: `USE_SUPABASE` |
| Deployment fails | Check Streamlit Cloud logs |

---

## 📚 Documentation Links

- **Main README**: Complete setup guide
- **SUPABASE_SETUP.md**: Database setup (step-by-step)
- **DEPLOYMENT.md**: Streamlit Cloud deployment
- **PROJECT_SUMMARY.md**: Full project overview
- **SUPABASE_INTEGRATION.md**: What changed with Supabase

---

## 🎓 Key Learnings

1. Streamlit for web UIs

2. LLM API integration (Gemini)
3. Prompt engineering
4. Database integration (Supabase)
5. Dual storage patterns
6. Cloud deployment
7. Environment management

---

## 💻 Development Workflow

```bash
# 1. Make changes
code app.py

# 2. Test locally
streamlit run app.py

# 3. Commit
git add .
git commit -m "Updated feature"

# 4. Push
git push

# 5. Auto-deploys to Streamlit Cloud! 🚀
```

---

## ⚡ Pro Tips

1. **Use Supabase** for production deployments
2. **Keep CSV** as backup (dual storage is smart)
3. **Test locally** before deploying
4. **Check logs** in Streamlit Cloud dashboard
5. **Monitor API usage** in Gemini/Supabase dashboards
6. **Document changes** in commit messages

---

## 📞 Support Resources

- **Streamlit**: docs.streamlit.io
- **Gemini**: ai.google.dev
- **Supabase**: supabase.com/docs
- **Project Docs**: Read all `.md` files in repo

---

## ✅ Success Checklist

Your app is ready when:

- ✅ Runs locally without errors
- ✅ AI responses generate correctly
- ✅ Data saves (CSV and/or Supabase)
- ✅ Recent feedback displays
- ✅ Deployed with public URL
- ✅ All tests pass
- ✅ Screenshots ready for report

---

**You're all set! Good luck with your assignment!** 🚀

*Last updated: December 2025*